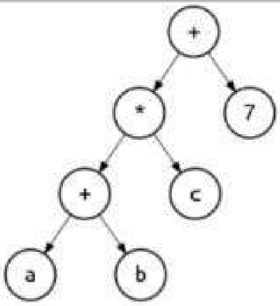
EAST WEST UNIVERSITY
Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Mid Term II Examination, Fall 2020 Semester

Course:	CSE 207- Data Structures, Section-1
Instructor:	Dr. Maheen Islam, Associate Professor, CSE Department
Full Marks:	40
Time:	1 Hour and 15 Minutes

1. A binary tree has eight nodes. The postorder and inorder traversals of the tree ar below. Draw the tree. Mark: 10

    Postorder: A6 A5 A3 A8 A7 A4 A2 A1
    Inorder: A6 A3 A5 A1 A2 A8 A4 A7

2. Given the following expression tree and the algorithm to find the correspo infix expression of the tree, show the recursive steps that are executed. Mark: 10




```
Algorithm infix (tree)
    if (tree not empty)
        if(tree token is an operand)
          print (tree token)
        else
          print ( open parenthesis)
          infix (tree left subtree)
          print (tree token)
          infix (tree right subtree)
          print (close parenthesis)
        end if
    end if;
```

3. Suppose we need to group a set of numbers while maintaining their origin order in each group. In this problem, we want to categorize the given set c numbers into four different groups: Mark: 10

    - Group 1: less than 10
    - Group 2: between 10 and 19
    - Group 3: between 20 and 29
    - Group 4: 30 and greater.

    To demonstrate, if the following number set is given: 3 22 12 6 10 34 65 2 30 81 4 5 19 20 57 44 99, after rearranging them into four groups it will be follows - Group-1: 3 6 9 4 5, Group-2: 12 10 19, Group-3: 22 29 20 and Group-4: 34 65 30 81 57 44 99. The result is not a sorted list but rather a l categorized according to the specified rules. The numbers in each group l kept their original order.
    Write a program that reads a set of numbers and categorizes them into appropriate groups according to the given rule. Here, you need to build a c for each of the four categories and store the numbers in the appropriate q as you read them. Finally, you must print each queue.

4. Write a function that will compute the sum of leaf nodes in each level of a l tree and will find the maximum among those. Mark: 10
    Considering the following example also show the steps executed by your pr function.

```
Input:
                1
               / \
             2     3
            / \     \
           4   5     8
                    / \
                   6   7
Output: 13
```